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3 March 1988

MILITARY SPECIFICATION
INSTALLATION KIT, ELECTRONIC EQUIPMENT, MK-2488/G

This specification is approved for use by the Communications Electronics Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of installation kit, designated Installation Kit, Electronic Equipment, MK-2488/G (See 6.1)

2. APPLICABLE DOCUMENTS.

2.1 Government Documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY	
MIL-P-116	Preservation-packaging, Methods of
MIL-S-901	Shock Test High Impact Shipboard Machinery Equipment
MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Cdr, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ. 07703-5000 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5815

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MIL-P-11268	Parts, Material, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-I-17563	Impregnants for Aluminum, Copper, Iron, Magnesium and Zinc alloy castings
MIL-I-45208	Inspection System Requirements
MIL-T-45542	Tool Set, Shop Sets and Kit Maintenance, Modifications and Tool Packaging of
MIL-P-55110	Printed Wiring Boards

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD 129	Marking For Shipment and Storage
MIL-STD-188-114	Electrical Characteristics of Digital Interface Circuits
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-810	Environmental Methods
MIL-STD-1275	Characteristics of 28 Volt DC Electrical Systems in Military Vehicles
MIL-STD-45662	Calibration System Requirement

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2.1.2 Other Government documents and drawings. The following other Government Documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

NATIONAL SECURITY AGENCY	
NACSIM 5100A	Compromising Emanations, Laboratory Test Requirements
DRAWINGS	
DLA3045888	Interconnecting Box, J-4024/U
DLA3045883	Cable Assembly, Power, Electrical CX-13315/G
DLA3045866	Cable Assembly, Special Purpose, Electrical, CX-13316/G
DLA3045882	Cable Assembly, Special Purpose, Electrical, CX-13317/G
DLA3045861	Mounting Base, Electrical Equipment, MT-6442/G
DLA3045884	Adapter, Signal, MX-10735/G
DLA3045860	Mounting Base, Intermediate MT-6445/VSC-2
DLA3045856	Mount,Resilient,MT-6444/VSC-3
A3045996	Power Supply Assembly
A3045994	Black Regulator Assembly
A3045997	Red Regulator Assembly
A3045993	Red Converter Assembly
A3045995	Black Converter Assembly

(Copies of the specification, standards, drawings and other Government documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number should be stipulated when requesting copies.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS.

3.1 First article. When specified in the procurement document the contractor shall furnish First Article units in accordance with 4.3

3.2 Construction. (See 4.4) Each of the components of the kit shall be in accordance with the requirements of this specification and the applicable drawings and data lists as follows:

Unit	Drawing List
Interconnecting Box, J-4024/U	DLA3045888
Mounting Base, Electrical Equipment, MT-6442/G	DLA3045861
Cable Assembly, Power, Electrical, CX-13315/G	DLA3045883
Cable Assembly, Special Purpose, Electrical, CX-13316/G	DLA3045886
Cable Assembly, Special Purpose, Electrical, CX-13317/G	DLA3045882

3.2.1 Ancillary equipment. (See 6.2b) If provided by the procurement document, the following items shall meet the requirements of this specification and applicable requirements of workmanship, inspection and service condition tests as specified herein.

3.2.1.1 Mount, resilient, MT-6444/VSC-3. The mount assembly shall be constructed in accordance with DLA3045856.

3.2.1.2 Mounting base, intermediate, MT-6445/VSC-2. The mounting base assembly shall be constructed in accordance with DLA3045860.

3.2.1.3 Adapter, signal, MX-10735/G. The adapter assembly shall be constructed in accordance with DLA3045884.

3.3 Parts, materials and processes, general. (See 4.4) In addition to the requirements of this specification, the requirements of MIL-P-11268, as specified herein shall apply:

Title	Paragraph No.
Applicability	3.2.2
Standard & Non-Standard Parts	3.3, 3.4.3-3.4.6,3.5,3.6
Materials	3.9-3.9.3.1,3.9.11, 3.9.13, 3.9.17

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Printed wiring Assemblies	3.10.32-3.10.35.3, 3.10.35.5-3.10.35.8
Semiconductor Devices	3.10.36
Processes	3.11-3.11.3,3.11.7, 3.11.9-3.11.11

3.3.1 Printed wiring. The printed wiring shall meet the requirements of MIL-P-55110.

3.3.2 Cable assemblies. Cable Assembly fabrication shall meet the requirements of this specification, applicable drawings and MIL-C-3885; 3.1,3.3-3.8.2.

3.3.3 Castings. Castings shall meet MIL-STD-454; Requirement 21.

3.3.4 Impregnation of aluminum castings. All Aluminum sand and die castings shall be impregnated in accordance with MIL-STD-276 using impregnation material per MIL-I-17563. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity.

3.4 Finish. (See 4.4) The equipment shall be finished in accordance with MIL-F-14072 and the drawings specified in 3.2.

3.5. Marking. (See 4.4, 4.13 and 4.14)

3.5.1 General. (See 4.13) Marking shall conform to MIL-M-13231.

3.5.2 Serial numbers. (See 4.14) Serial numbers shall be placed on the Interconnecting Box, J-4024/U of the MK-2488/G Installation Kit.

3.6 Interchangeability. (See 4.11) Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454.

3.7 Electrical requirements. (See 4.6-4.7)

3.7.1 J-4024/U interconnecting box.

3.7.1.1 Red signal interfaces.

3.7.1.1.1 Red transmit input signal interface. The red transmitter on the Red Converter Assembly shall accept neutral, normal input keying signals, mark on, at information rates of 50 and 75 baud from the following teletypewriter sources:

Communications Terminal, AN/UGC-74A(V)3, 70 microamp mode only

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Teletypewriter Set, TT-4(*)/TG
Teletypewriter Set, TT-98(*)/TG
Teletypewriter Reperforator-Transmitter TT-76(*)/GGC

* Indicates modifications of the specified set.

The steady state current provided by the red transmitter input circuit to closed keying contacts at the J-4024/U red transmitter input connector shall be 50 to 70 microamps. For the purpose of testing, steady state and dynamic red transmitter input signals shall be provided to the J-4024/U through input circuits equivalent to the teletypewriter keyboard and transmitter distributor, and/or the transmit output of the communications terminal specified above.

3.7.1.1.2 Red transmit output signal interface. The red Transmitter on the Red Converter Assembly shall produce a low level digital signal in accordance with MIL-STD-188-114, positive mark, for the teletypewriter input signals specified in 3.7.1.1.1. The red transmit output signal shall have less than 1% bias and less than +/- 1% end distortion of the red transmit input signal for information rates of 50 and 75 baud.

3.7.1.1.3 Red receiver input signal interface. The red receivers on the Red Converter Assembly shall accept low level polar signals conforming to MIL-STD-188-114 positive mark for information rates of 50 and 75 baud.

3.7.1.1.4 Red receiver output signal interface. Each of the two red receivers on the Red Converter Assembly shall produce neutral, 20ma -1/+2ma, mark on, impulse at information rates of 50 and 75 baud to operate the following teletypewriter sets:

Communication Terminal AN/UGC-74A(V)3 in 48V receive mode only
Teletypewriter Set TT-4(*)/TG
Teletypewriter set TT-98(*)/TG
Teletypewriter Reperforator-Transmitter TT-76(*)/GGC

* Indicates modifications of the specified set.

J-4024/U red receiver output signals shall meet steady state and dynamic signal requirements of the test in 4.7.5 and 4.7.6 when the red receiver circuit output interfaces are loaded with circuits equivalent to the selector magnets of the teletypewriters and/or the receiver input circuit of the communications terminal specified above. Dynamic signal inputs to the J-4024/U red receivers shall be in accordance with 3.7.1.1.3 at the red receiver input

3.7.1.1.5 Red receiver input signal capacity. The J-4024/U shall have provisions for driving both of the red receivers on the Red Converter Assembly simultaneously with signals conforming to MIL-STD-188-114 at information rates of 50 and 75 baud from an external source device.

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Provisions shall also be made for driving either one of the red receivers with the red transmitter output signal (Home Copy function) while simultaneously driving the other red receiver with signals from an external source device. Manual selection from the J-4024/U front panel shall determine the source of signals for the red receivers.

3.7.1.1.6 Red receiver output signal capacity. The J-4024/U shall provide red output signal interfaces for connection of any combination of two of the teletypewriter sets and/or communications terminal specified in 3.7.1.1.4.

3.7.1.1.7 J-4024/U Red signal to TSEC/KG-84A interface. The J-4024/U shall provide for connection of the red transmitter output to the TSEC/KG-84A in accordance with MIL-STD-188-114 balanced generator to unbalanced receiver requirements. The J-4024/U shall provide (in the same connector) for connection of the red receiver input from the TSEC/KG-84A in accordance with MIL-STD-188-114 unbalanced generator to balanced receiver requirements. Provisions for electrical grounding at the signal interfaces shall be in accordance with MIL-STD-188-114.

3.7.1.2 Black signal interfaces.

3.7.1.2.1 Black transmit input signal interface. The black transmitter on the Black Converter Assembly shall accept low level, digital, polar signals conforming to MIL-STD-188-114.

3.7.1.2.2 Black transmit output signal interface. The black transmitter on the Black Converter Assembly shall provide a closed circuit output in response to a mark signal at the input to the black transmitter or an open circuit output in response to a space signal at the input of the black transmitter at information rates of 50 and 75 baud. The J-4024/U black transmitter output signals shall operate a 20 +/-2 milliamp current loop having a power source voltage of 60 VDC to 130 VDC as provided by the following modems:

Modem MD-522(*)/GRC
Terminal, Telegraph TH-22(*)/TG
Terminal, Telegraph TH-5(*)/TG

* Indicates modifications of the specified modem.

The black transmitter on the Black Converter Assembly shall provide for normal operation of the specified modems with normal or reversed connections between the J-4024/U and the modem. J-4024/U black transmitter output signals shall meet the steady state and dynamic signal requirements of the tests in 4.7.5 and 4.7.6 when the black transmitter output interface is loaded with a circuit equivalent to the send input circuit of the modems specified above. Dynamic signal inputs to the black transmitter shall be in accordance with 3.7.1.2.1 at the black transmitter input.

3.7.1.2.3 Black receive input signal interface. The receiver on the Black Converter Assembly shall accept neutral, 20 +/- 2 milliamp impulses, mark on, at information rates of 50 and 75 baud from current

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loops having a power source voltage of 127 VDC to 280 VDC as provided by the modems specified in 3.7.1.2.2. The black receiver on the Black Converter Assembly shall provide for normal operation of the J-4024/U with normal or reversed connections between the modem and the J-4024/U. For the purpose of testing, black receiver input signals shall be provided from a circuit equivalent to the receive output circuits of the modems specified in 3.7.1.2.2.

3.7.1.2.4 Black receive output signal interface. The black receiver on the Black Converter Assembly shall provide low level, polar, digital signals in accordance with MIL-STD-188-114 at information rates of 50 and 75 baud in response to the input signal specified in 3.7.1.2.3.

3.7.1.2.5 J-4024/U to TSEC/KG-84A interface. The J-4024/U shall provide for connection of the black transmitter input from the TSEC/KG-84A in accordance with MIL-STD-188-114 unbalanced generator to balanced receiver requirements. The J-4024/U shall provide (in the same connector) for connection of the black receiver output to the TSEC/KG-84A in accordance with MIL-STD-188-114 unbalanced generator to unbalanced receiver requirements. Electrical grounding provisions at the signal interfaces shall be in accordance with MIL-STD-188-114.

3.7.1.2.6 J-4024/U to modem interface. The J-4024/U shall provide for two wire connection of the black transmitter output to the modem and two wire connection to the black receiver input from the modem. Connectors on the J-4024/U for the connectors to the modem shall be banana jacks.

3.7.1.3 J-4024/U power interfaces.

3.7.1.3.1 Input power sources. The J-4024/U shall operate on any of the following input power sources:

115 VAC +/- 10%, 47 to 63 Hz, single phase
230 VAC +/- 10%, 47 to 63 Hz, single phase
28 VDC (nominal) +4,-6 VDC

The input power source shall be manually selectable on the rear panel such that the J-4024/U and the TSEC/KG-84A shall operate only if the input power source electrical parameters (correctly presented to the J-4024/U input power connectors) and input power source selector parameters are matched and the power switches on the J-4024/U and TSEC/KG-84A are in the on position.

3.7.1.3.2 J-4024/U power source interface capacity. The J-4024/U shall provide for connection of an AC power source and a DC power source simultaneously.

3.7.1.3.3 J-4024/U to TSEC/KG-84A power interface. The J-4024/U shall provide interruptible AC or DC input power to the TSEC/KG-84A in accordance with the input power source with out recabling the J-4024/U

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to the TSEC/KG-84A interface. Power to the TSEC/KG-84A shall be provided only when the J-4024/U power is on.

3.7.1.3.4 Overload. Overload protection shall be in accordance with applicable requirements 1 and 8 (Class 1) of MIL-STD-454.

3.7.1.3.5 Overvoltage and transient. The J-4024/U shall operate within the fault free parameters of mobile 28VDC power sources as specified in MIL-STD-1275.

3.7.1.3.6 Reverse polarity. When connected to DC power the J-4024/U shall not operate if the polarity of the DC power source connection to the J-4024/U is incorrect.

3.7.2 Cable assembly, power, electrical, CX-13315/G and cable assembly, special purpose electrical, CX-13316/G and CX-13317/G. (See 4.7.1) The Cable Assemblies shall meet the wiring requirements of the appropriate drawings (DLA3045883, DLA3045866, and DLA3045882). Conductors in the cable assembly shall be continuous and have pin to pin resistance of less than 1 ohm. When tested for dielectric strength in accordance with 4.8.5.1 there shall be no arc over or breakdown at 1000 VDC. When tested in accordance with 4.8.5.1 at 500 VDC the insulation resistance shall be at least 100 megohms.

3.7.3 Adapter, signal, MX-10735/G. (See 4.7.2) The Signal Adapter shall meet the wiring requirements of DLA3045884. Conductors shall be continuous and shall have a pin to pin resistance of less than 1 ohm.

3.7.4 J-4024/U harness assemblies. (See 4.7.3) Harness assembly conductors shall be continuous from connection point to connection point and have a point-to-point resistance of less than 1 ohm as measured in 4.7.4.1.

3.8 Service conditions. The MK-2488/G component assemblies shall be subjected to tests of service conditions as specified below. The MK-2488/G shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered; no movable part or control shall be shifted in setting, position, or adjustment; and no permanent degradation shall be caused in performance below that specified in 3.7 as determined by the MK-2488/G Post EV test (4.7.6.2), when subjected to the following environmental conditions.

3.8.1 Installation kit, electronic equipment, MK-2488/G; mounting base, intermediate, MT-6445/VSC-2; mount, resilient, MT-6444/VSC-3; and adapter, signal, MX-10735/G.

3.8.1.1 Temperature.

3.8.1.1.1 Low temperature. (See 4.8.1.1.1) During test the MK-2488/G shall meet the Normal operational verification portion of the MK-2488/G Post EV test, 4.7.6.2. After the test the equipment shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2: each

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Cable Assembly shall meet the requirements of 3.7.2; and the Signal Adapter shall meet the requirements of 3.7.3.

3.8.1.1.2 High temperature. (See 4.8.1.1.2) During the test the MK-2488/G shall meet the Normal operation verification portion of the MK-2488/G Post EV test, 4.7.6.2. After the test the equipment shall meet the requirements of the MK-2488/G Post EV test, 4.7.6.2: each Cable Assembly shall meet the requirements of 3.7.2; and the Signal Adapter shall meet the requirements of 3.7.3.

3.8.1.2 Low pressure. (See 4.8.1.2) During the test the MK-2488/G shall meet the Normal electrical operation verification portion of the MK-2488/G Post EV test 4.7.6.2. After the test the equipment shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2; each Cable Assembly shall meet the requirements of 3.7.2; and the Signal Adapter shall meet the requirements of 3.7.3.

3.8.1.3 Humidity. (See 4.8.1.3) During the test the MK-2488/G shall meet the Normal electrical operation verification portion of the MK-2488/G Post EV test, 4.7.6.2. After the test the equipment shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2; each Cable Assembly shall meet the requirements of 3.7.2; and the Signal Adapter shall meet the requirements of 3.7.3.

3.8.1.4 Fungus. (See 4.8.1.4) After the test the equipment and assemblies thereof shall show no more than sparse microbial growth with restricted tuberal growth (See 6.8.1-6.8.6) development in an area 10% less of the total area and no more than six unrelated minute colonies with mycelial development in noncritical areas. Critical areas include terminal spacing of printed circuit boards. After the test the equipment shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2; each Cable Assembly shall meet the requirements of 3.7.2; and the Signal Adapter shall meet the requirements of 3.7.3.

3.8.1.5 Sand and dust. (See 4.8.1.5) During the test the MK-2488/G shall meet the Normal electrical operation verification portion of the MK-2488/G Post EV test, 4.7.6.2; each Cable Assembly shall meet the requirements of 3.7.2; and the Signal Adapter shall meet the requirements of 3.7.3.

3.8.2 Interconnecting box, J-4024/U; mounting base, electrical equipment, MT-6442/G; mounting base, intermediate, MT-6445/VSC-2; and mount, resilient, MT-6444/VSC-3.

3.8.2.1 Salt fog. (See 4.8.2.1) After the test the equipment when visually examined shall show no evidence of degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface or exfoliation (See 6.8.7) of metal. In addition, the equipment shall exhibit no evidence of clogging or binding when an attempt is made to operate any mechanical part through its full range of movement. After the test

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the equipment (with Cable Assemblies that have been separately tested for salt fog conditions) shall meet all the requirements of MK-2488/G Post EV test, 4.7.6.2.

3.8.2.2 Immersion. (See 4.8.2.2) After the test there shall be no evidence of leakage in the J-4024/U in the internal VMI steps of the MK-2488/G Post EV test, 4.7.6.2. After the test the equipment (with Cable Assemblies that have been separately tested for immersion conditions) shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2.

3.8.3 Installation kit, electronic equipment, MK-2488/G; and adapter, signal, MX-10735/G.

3.8.3.1 Vibration. (See 4.8.3.1) After the test the MK-2488/G shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2; each Cable Assembly shall meet the requirements of 3.7.2; and the Signal Adapter shall meet the requirements of 3.7.3.

3.8.3.2 Shock. (See 4.8.3.2) After the test the MK-2488/G shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2; each Cable Assembly shall meet the requirements of 3.7.2; and the Signal Adapter shall meet the requirements of 3.7.3.

3.8.4 Installation kit, electronic equipment, MK-2488/G; and mount, resilient, MT-6444/VSC-3.

3.8.4.1 Vibration. (See 4.8.4.1) After the test the equipment shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2; and each Cable Assembly shall meet the requirements of 3.7.2.

3.8.4.2 Ballistic shock. (See 4.8.4.2) After the test the equipment shall meet all the requirements of the MK-2488/G Post EV test, 4.7.6.2; and each Cable Assembly shall meet the requirements of 3.7.2.

3.8.5 Cable assembly, power, electrical, CX-13315/G and cable assemblies, special purpose, electrical, CX-13316/G and CX-13317/G.

3.8.5.1 Axial load. (See 4.8.5.1) Each Cable Assembly shall withstand loads of 250 +/- 5 lbs without evidence of permanent physical damage. At the time of maximum load, the cable assembly conduit shall not experience an elongation exceeding fifteen (15) percent of the dormant length. There shall be no mechanical failure of any of the component parts. After the test the Cable Assembly shall meet the requirements of 3.7.2.

3.8.5.2 Salt fog. (See 4.8.5.2) After testing the Cable Assemblies, when visually examined, shall show no evidence of scratches, abrasions, breaks, blistering, or otherwise damaged components. In addition, the equipment shall exhibit no evidence of clogging or binding when an attempt is made to operate the connector backshell

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assemblies. After the test the Cable Assemblies shall meet the requirements of 3.7.2.

3.8.5.3 Immersion. (See 4.8.5.3) After the test there shall be no evidence of leakage in the Cable Assemblies. After the test the Cable Assemblies shall meet the requirements of 3.7.2.

3.8.5.4 Cold bend. (See 4.8.5.4) After the test the Cable Assemblies shall meet the requirements of 3.7.2.

3.9 Electromagnetic interference. (See 4.9) The MK-2488/G shall meet the emissions and susceptibility requirements of MIL-STD 461:

- CE01
- CE02
- CE03
- CE04
- CE05
- CS01
- CS02
- CS06
- RE02
- RE02.1
- RS02
- RS03*

* The electric field intensities and modulation characteristics for RS03 shall be as follows:

FREQUENCY RANGE	FIELD INTENSITY	MODULATION CHAR.
10 Khz to 2 Mhz	1 volt/meter	AM, 50%, 1000 Hz Tone
2 Mhz to 30 Mhz	5 volt/meter	AM, 50%, 1000 Hz Tone
30 Mhz to 76 Mhz	10 volt/meter	FM,+/-1Khz Dev,1000Hz Tone
76 Mhz to 400 Mhz	10 volt/meter	AM, 50%, 1000 Hz Tone
400 Mhz to 2 Ghz	10 volt/meter	Pulse,1usec pulse width, 800PPS

3.10 Tempest. (See 4.10) The MK-2488/G shall meet the requirements of NACSIM 5100A.

3.11 Unit subassembly electrical performance. (See 4.6) The unit subassemblies listed below shall meet the requirements of specifications described in 3.7 and their respective test requirement drawings as follows:

Power Supply Assembly	A3045996
Black Regulator Assembly	A3045994
Red Regulator Assembly	A3045997
Red Converter Assembly	A3045993
Black Converter Assembly	A3045995

3.12 Electrical preconditioning (burn-in). (See 4.7.5) After all subassemblies VMI and electrical performance tests, and assembly VMI and continuity tests, each assembled J-4024/U Interconnecting Box shall be electrically preconditioned for a minimum of 12 hours all of which shall be failure free. Failure free is defined as (1) no loss of signal as specified in 4.7.5 during the burn in period and (2) sucessful completion of the Electrical operation verification test of 4.7.6.1.

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3.13 Operation verification tests. (See 4.7.6) Each fabricated and tested unit shall be subjected to operation verification tests in accordance with 4.7.6 after all other inspections and tests are completed to determine that the unit is operating prior to packaging.

3.14 System safety engineering. (See 4.12)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with the requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, and lenses).

3.15 Workmanship. (See 4.2) Workmanship of the equipment manufactured and assembled under the procurement document shall be in accordance with Requirement 9 of MIL-STD-454, applicable portion of MIL-P-11268 (see 3.3), and this specification in 3.3.1-3.3.4, 3.4 and 3.5.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the procurement document, the contractor is responsible for the performance of all inspection requirements as specified herein. The contractor's inspections system shall be in accordance with MIL-I-45208 and MIL-STD-45662. Except as otherwise specified in the procurement document, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (See 4.3): does not include packaging.

b. Inspections covered by subsidiary documents (See 4.4).

c. Quality conformance inspections of equipment before packaging (See 4.5).

d. Packaging inspection (See 4.15).

4.3 First article. (See 3.1) Unless otherwise specified in the procurement document the inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of eight (8) First Article units of the complete Installation Kit, Electronic Equipment, MK-2488/G, each with a Mounting Base, Intermediate, MT-6445/VSC-2; Mount, Resilient, MT-6444/VSC-3; and Adapter, Signal, MX-10735/G, unless otherwise specified in the procurement document.

4.3.2 First article inspection. The First Article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

TABLE I. First article inspection

INSPECTIONS	RQMT PARA	TEST PARA	UNIT 1	2	3	4	5	6	7	8
1. Inspections by Subsidiary Documents (Note a)	3.14 3.15	4.4	Inspection to be performed on all units							
2. Group A Inspection	See Table II		Inspection to be performed on all units							
3. Group B Inspection	See Table III		Inspection to be performed on all units							
4. Group C Inspection										
Installation kit, electronic equipment, MK-2488/G; mounting base, intermediate, MT-6445/VSC-2; mount, resilient, MT-6444/VSC-3; adapter, signal, MX-10735/G					(note b)					
Low Temperature	3.8.1.1.1	4.8.1.1.1	1							
High Temperature	3.8.1.1.2	4.8.1.1.2	2							
Low Pressure	3.8.1.2	4.8.1.2	3							
Humidity	3.8.1.3	4.8.1.3						1		
Fungus (Note c)	3.8.1.4	4.8.1.4		1						

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TABLE I. First article inspection (Cont.)

INSPECTIONS	RQMT PARA	TEST PARA	UNIT 1	2	3	4	5	6	7	8
Sand and Dust (Note c)	3.8.1.5	4.8.1.5			1					
Interconnecting box J-4024/U; mounting base, electrical equipment, MT-6442/G; mounting base, intermediate, MT-6445/VSC-2; mount, resilient, MT-6444/VSC-3										
Salt Fog (Note c)	3.8.2.1	4.8.2.1				1				
Immersion	3.8.2.2	4.8.2.2				2				
Installation kit, electronic equipment, MK-2488/G; and adapter, signal MX-10735/G										
Vibration (Note d)	3.8.3.1	4.8.3.1					1			
Shock (Note d)	3.8.3.2	4.8.3.2					2			
EMI/EMC (Note e)	3.9	4.9							1	
TEMPEST (Note e)	3.10	4.10							2	
Installation kit, electronic equipment, MK-2488/G; and mount, resilient, MT-6444/VSC-3										
Vibration (Note d)	3.8.4.1	4.8.4.1								1
Ballistic Shock (Note d)	3.8.4.2	4.8.4.2								2
Cable assembly, power, electrical, CX-13315/G; and cable assembly, special purpose, electrical, CX-13316/G and CX-13317/G										
Salt Fog (Note c)	3.8.5.2	4.8.5.2				3				
Immersion	3.8.5.3	4.8.5.3				4				
Axial Load	3.8.5.1	4.8.5.1	5			5		2		
Cold Bend	3.8.5.4	4.8.5.4	6			6		3		

NOTES

a. Inspection 1 through 3 shall be performed in the order shown on all First Article Units before subjecting these units to any other inspection requirements specified in the table. Subsidiary Documents Inspections that are covered in Group A Inspections shall be performed only once as a part of Item 2.

b. The numbers in the columns for Group C Inspections (Item 4) specify the order of inspection for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned dried, and refurbished after this inspection before proceeding with subsequent

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inspections.

d. Tests (except for bench handling shock) shall include a dummy box of similar size and weight as the TSEC/KG-84A with dummy connectors for the Cable Assemblies (and Signal Adapter, if required).

e. Tests shall be conducted without Signal Adapter.

4.3.3 First article data. The First Article Test Plan and Test Report(s) shall be as required in the procurement document.

4.4 Inspection covered by subsidiary documents. The following items shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the procurement document.

Item	Where Required
Parts, Materials, and Processes	3.3
Dimensional Verification	3.6
Printed Wiring	3.3.1
Cable Assemblies	3.3.2
Impregnation and quality of casting	3.3.4
Finish	3.4
Marking	3.5
Workmanship	3.15

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit (and each subassembly identified in Table II) in the procurement document shall be inspected for conformance to the inspections specified in Table II (Group A Inspection). Units shall be compromised of subassemblies which have been subjected to and passed the Group A Inspection and Quality Assurance Verification cited herein. Discrete lots for Quality Assurance Verification shall be formed of units and subassemblies that passed the Group A Inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be submitted to sampling inspection, utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table II (Quality Assurance Verification). Group A inspection shall be performed in any order which is satisfactory to the Government, except that the Electrical operation verification test, 4.7.6, shall be last. The contractor shall specify the order of Group A inspections in the First Article Test Plan.

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TABLE II. Group A inspection

INSPECTION	REQMT PARA	INSP PARA	AQL MAJOR	AQL MINOR
VISUAL-MECHANICAL (Including Spare Assemblies)				
J-4024/U Printed Wiring Boards				
Power Supply Assy	3.15	4.12	2.5%	4.0%
Red Regulator Assy	3.15	4.12	2.5%	4.0%
Black Regulator Assy	3.15	4.12	2.5%	4.0%
Red Converter Assy	3.15	4.12	2.5%	4.0%
Black Converter Assy	3.15	4.12	2.5%	4.0%
J-4024/U Harness Assemblies				
Red Harness Assy w/Filter	3.15	4.12	2.5%	4.0%
Black Harness Assy	3.15	4.12	2.5%	4.0%
Power Harness Assy w/Filters	3.15	4.12	2.5%	4.0%
Red Power Cable	3.15	4.12	2.5%	4.0%
J-4024/U Interconnecting Box				
J-4024/U Housing Assy	3.15	4.12	2.5%	4.0%
J-4024/U Cover Assy	3.15	4.12	2.5%	4.0%
J-4024/U Back Plate Assy	3.15	4.12	2.5%	4.0%
Power Supply Assy	3.15	4.12	2.5%	4.0%
J-4024/U w/PWBs & Harnesses	3.15	4.12	2.5%	4.0%
Cable Assemblies				
Electrical Power Cable Assy CX-13315/G	3.15	4.12	2.5%	4.0%
Special Purpose Cable Assy CX-13316/G	3.15	4.12	2.5%	4.0%
Special Purpose Cable Assy CX-13317/G	3.15	4.12	2.5%	4.0%
Signal Adapter MX-10735/G	3.15	4.12	2.5%	4.0%
Mounts				
Electrical Equipment Mounting Base MT-6442/G	3.15	4.12	2.5%	4.0%
Intermediate Mounting Base MT-6445/VSC-2	3.15	4.12	2.5%	4.0%
Resilient Mount MT-6444/VSC-3	3.15	4.12	2.5%	4.0%
ELECTRICAL PERFORMANCE				
J-4024/U Printed Wiring Boards				
Power Supply Assy	3.11	4.6.1	1.0%	*
Black Regulator Assy	3.11	4.6.2	1.0%	*
Red Regulator Assy	3.11	4.6.3	1.0%	*
Black Converter Assy	3.7.1.2.1-.4;3.11	4.6.4	1.0%	*
Red Converter Assy	3.7.1.1.1-.4;3.11	4.6.5	1.0%	*

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TABLE II. Group A inspection (Cont.)

INSPECTION	REQMT PARA	INSP PARA	AQL MAJOR	AQL MINOR
Power Supply Assembly	3.7.1.3.1	4.6.6	1.0%	*
J-4024/U Harness Assemblies				
Red Harness Assy	3.7.4	4.7.2	1.0%	*
Black Harness Assy	3.7.4	4.7.2	1.0%	*
Power Harness Assy w/Filters	3.7.4	4.7.2	1.0%	*
Cable Assemblies				
Electrical Power Cable Assy CX-13315/G	3.7.2	4.7.3	1.0%	*
Special Purpose Cable Assy CXS-13316/G	3.7.2	4.7.1	1.0%	*
Special Purpose Cable Assy CX-13317/G	3.7.2	4.7.1	1.0%	*
Signal Adapter MX-10735/G	3.7.3	4.7.3	1.0%	*
J-4024/U Interconnecting Box				
Continuity	3.7.1	4.7.4	1.0%	*
Power Up	3.7.4	4.7.4	1.0%	*
Burn-In	3.12	4.7.6	1.0%	*
Operation Verification	3.13	4.7.6	1.0%	*

* All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in Table III and the inspection level shall be S-4.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in the order listed in Table III.

TABLE III Group B inspections

INSPECTION	REQMT PARA	INSP PARA	AQL
MECHANICAL INTERCHANGEABILITY	3.6	4.5.2,4.11	6.5%
INTERCONNECTING BOX J-4024/U WITH:			
Mounting Base, Electrical Equipment, MT-6442/G			
Mount, Resilient, MT-6444/VSC-3			
Mounting Base, Intermediate, MT-6445/VSC-2			
Cable Assembly, Power, Electrical CX-13315/G			

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TABLE III Group B inspections

INSPECTION	REQMT PARA	INSP PARA	AQL
Cable Assembly, Special Purpose, Electrical, CX-13316/G			
Cable Assembly, Special Purpose, Electrical, CX-13317/G			
WITHIN INTERCONNECTING BOX, J-4024/U			
Power Supply Assembly			
Red Converter Assembly			
Red Regulator Assembly			
Black Converter Assembly			
Black Regulator Assembly			
ELECTRICAL INTERCHANGEABILITY	3.6	4.5.2	*
INTERCONNECTING BOX, J-4024/U WITH:			
Cable Assembly, Power, Electrical, CX-13315/G			
Cable Assembly, Special Purpose, Electrical, CX-13316/G			
Cable Assembly, Special Purpose, Electrical, CX-13317/G			
WITHIN INTERCONNECTING BOX, J-4024/U			
Power Supply Assembly			
Red Converter Assembly			
Red Regulator Assembly			
Black Converter Assembly			
Black Regulator Assembly			

* All electrical defects are major.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV for the production unit quantities specified in Table IV. Samples shall be selected in accordance with 4.5.3.1 except that for quantities of less than 100 the First Article Group C tests shall be the only Group C tests required.

TABLE IV Group C inspections

INSPECTION	INSPECTION REQ. IF PRODUCTION QUANTITY			REQUIRED PARAGRAPH	INSPECTION PARAGRAPH
	100-299	300-499	500+		
Subgroup 1					
Low Temperature*	Y	Y	Y	3.8.1.1.1	4.8.1.1.1
High Temperature	Y	Y	Y	3.8.1.1.2	4.8.1.1.2
Low Pressure	Y	Y	Y	3.8.1.2	4.8.1.2
Humidity	Y	Y	Y	3.8.1.3	4.8.1.3
Axial Load(CA only)	Y	Y	Y	3.8.5.1	4.8.5.1

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TABLE IV Group C inspections

INSPECTION	INSPECTION REQ. IF PRODUCTION QUANTITY 100-299	300-499	500+	REQUIRED PARAGRAPH	INSPECTION PARAGRAPH
Subgroup 2					
Vibration (MK-2488/G)	N	Y	Y	3.8.3.1	4.8.3.1
Shock (MK-2488/G)	N	Y	Y	3.8.3.2	4.8.3.2
EMI (MK-2488/G)	N	Y	Y	3.9	4.9
TEMPEST (MK-2488/G)	N	Y	Y	3.10	4.10
Subgroup 3					
Sand and Dust	N	N	Y	3.8.1.5	4.8.1.5**
Cold Bend (CAs only)	N	Y	Y	3.8.5.4	4.8.5.4
Subgroup 4					
Salt Fog (w/o CAs)	N	Y	Y	3.8.2.1	4.8.2.1**
Salt Fog (CAs only)	N	Y	Y	3.8.5.2	4.8.5.2**
Immersion (w/o CAs)	N	Y	Y	3.8.2.2	4.8.2.2
Immersion (CAs only)	N	Y	Y	3.8.5.3	4.8.5.3
Fungus	N	N	Y	3.8.1.4	4.8.1.4**
Vibration (w/MT-6444/VSC-3)	N	N	Y	3.8.4.1	4.8.4.1
Ballastic Shock (w/MT-6444/VSC-3)	N	N	Y	3.8.4.2	4.8.4.2

Y = Yes N = No CA = Cable Assemblies

* Unless otherwise stated tests are applicable to the ancillary equipment (as specified in 4.8) if provided by the procurement document.

** The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.5.3.1 Sampling for group C inspection. If the procurement document specifies ancillary items they shall be included in the samples for subsequent Group C tests as specified below and tested as required in 4.8.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot for First Article testing (See Table I for test order). For subsequent Group C inspection, one unit shall be selected every third month in accordance with the production quantity requirements of Table IV. The unit shall be subjected to the tests in the order shown in Table IV for Subgroup 1.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot for First Article testing (See Table I for test order). For subsequent Group C inspection, one unit shall be selected every fourth month and one unit shall be selected every eighth month in accordance with the production quantity requirements

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of Table IV. The unit selected every fourth month shall be subjected to vibration and shock (MK-2488/G only) testing in that order. The unit selected every eighth month shall be subjected to EMI/EMC and TEMPEST testing.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot for First Article testing (See Table I for test order). For subsequent Group C inspection, one unit shall be selected every fifth month in accordance with the production quantity requirements of Table IV. The unit shall be subjected to testing listed in Table IV for Subgroup 3.

4.5.3.1.4 Subgroup 4. For this subgroup, four units shall be selected from the first production lot for First Article testing (See Table I for test order). For subsequent Group C inspection four units shall be selected every twelfth month in accordance with the production quantity requirements of Table IV. One unit shall be selected for salt fog testing, one unit shall be selected for fungus testing, one unit shall be selected for immersion testing and one unit shall be selected for vibration and ballastic shock testing.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in the order specified in 4.5.3.1 and Table IV in accordance with the production quantity requirements of Table IV.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the procurement document (See 6.5).

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Noncompliance. If a sample fails to pass group C or D inspections, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of the product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C or D inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C or D inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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4.6 Subassembly electrical performance tests. (See 3.11) Subassemblies shall be subjected to electrical performance tests specified herein.

4.6.1 Power supply assembly. Each Power Supply Assembly shall be subjected to testing in accordance with A3045996.

4.6.2 Black regulator assembly. Each Black Regulator Assembly shall be subjected to testing in accordance with A3045994.

4.6.3 Red regulator assembly. Each Red Regulator Assembly shall be subjected to testing in accordance with A3045997.

4.6.4 Black converter assembly. Each Black Converter Assembly shall be subjected to testing in accordance with A3045995.

4.6.5 Red converter assembly. Each Red Converter Assembly shall be subjected to testing in accordance with A3045993.

4.7 Electrical tests.

4.7.1 Cable assembly electrical tests. (See 3.7.2) Each Cable Assembly shall be tested for continuity between corresponding connector pins as specified in the applicable drawings. Each Cable Assemble shall be tested to verify that an open circuit exists between each cable assembly conductor in turn, and any other conductor or connector shell that according to the applicable drawing is not connected to the conductor under test. A suitable digital multimeter for reading 0.5 ohms +/- 1 percent or an equivalent measurement instrument may be used.

4.7.2 Signal adapter electrical tests. (See 3.7.3) The Signal Adapter shall be tested for continuity between corresponding connector pins as specified in the applicable drawing. The Signal Adapter shall be tested to verify that an open circuit exists between each adapter assembly conductor in turn, and any other conductor or connector shell that, according to the applicable drawing, is not connected to the conductor under test.

4.7.3 Harness assemblies electrical tests. (See 3.7.4) Each Harness Assembly shall be tested for continuity between corresponding connector pins and conductors specified in the applicable drawing. Each Harness assembly shall be tested to verify that an open circuit exists between each Harness Assembly conductor in turn, and any other conductor or connector shell that, according to the applicable drawing, is not connected to the conductor under test.

4.7.4 Interconnecting box, J-4024/U electrical tests. Each J-4024/U shall be tested for continuity prior to electrical preconditioning as follows:

4.7.4.1 J-4024/U continuity test. (See 3.7.4) Each J-4024/U shall

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be tested for wiring continuity as follows:

a. Remove the J-4024/U Cover Assembly.

b. Disconnect and remove the Power Supply Assembly, Red Regulator Assembly, Black Regulator Assembly, Red Converter Assembly and Black Converter Assembly from the Interconnecting Box.

c. Connect the J15 continuity test plug on P15 and the J9 continuity test plug on P9 in the black compartment of the Interconnecting Box (See Figure 1).

d. Connect the J16 continuity test plug on P16 and the J10 continuity test plug on P10 in the red compartment of the Interconnecting Box (See Figure 1).

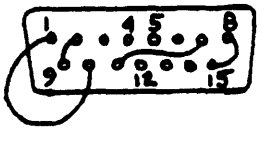
e. Set Interconnecting Box switches in positions shown on Table V and with suitable digital multimeter check continuity between Low TP and High TP as shown in Table V.

f. Set Interconnecting Box Switches in positions shown in Table VI and with suitable digital multimeter check continuity between Low TP and High TP as shown in Table VI.

g. Set Interconnecting Box switches in positions shown in Table VII and with a suitable digital multimeter check continuity between Low TP and High TP as shown in Table VII.

h. Connect the digital multimeter (DMM) between J15-7 in the black compartment (black regulator assembly) and DS1-2 (power indicator lamp) in the power compartment. The DMM shall indicate less than .1 ohm.

i. Connect the DMM between P13-A and DS1-1 in the power compartment. The DMM shall indicate less than .1 ohm.

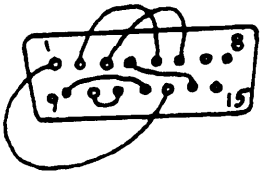


JUMPERS	
FROM PIN	TO PIN
1	10
2	9
7	11
8	15

J9 TEST PLUG

FIGURE 1 Continuity test plugs

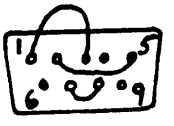
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JUMPERS

FROM PIN	TO PIN
1	13
2	5
3	6
4	14
9	12
10	11

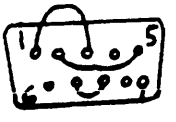
J10 TEST PLUG



JUMPERS

FROM PIN	TO PIN
1	3
2	5
7	8

J15 TEST PLUG



JUMPERS

FROM PIN	TO PIN
1	3
2	5
7	8

J16 TEST PLUG

FIGURE 1 Continuity test plugs (Cont.)

TABLE V Interconnecting box, J-4024/U black compartment continuity test

J-4024/U switch settings: S1: OFF, S2: 115, S3: 1,S4: OFF

LOW TP	HIGH TP															
	<-----J1---->					<-J9->		<---P13-->				<---Banana Jack-->				
	A	B	C	D	F	12	14	A	B	D	E	L1	L2	L3	L4	L5
J1A	-	X	X	X	X	X	X	-	X	X	X	X	C	X	X	X
J1B	X	-	X	X	X	X	X	-	X	X	X	X	X	X	C	X
J1C	X	X	-	X	X	X	X	-	X	X	C	X	X	C	X	X
J1D	X	X	X	-	X	X	X	-	X	X	X	C	X	X	X	X
J1F	X	X	X	X	-	X	X	-	X	X	X	X	X	X	X	C
P13B	X	X	X	X	X	X	C	-	-	X	X	X	X	X	X	X
P13D	X	X	X	X	X	C	X	-	X	-	X	X	X	X	X	X
P13E	X	X	C	X	X	X	X	-	X	X	-	X	X	C	X	X

X = open circuit C = less than or equal to .1 ohm
- = no test TP = test probe

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TABLE VI Interconnecting box, J-4024/U, power compartment continuity test.

J-4024/U Switches S1	S2	Low TP	High TP	(---J3---) A	B	C	D	(-J11-) A	B	C	(----------J4----------) A	B	C	D	E	H	J	K	L	M	N	(----------J12----------) A	B	C	D	E	F	H	J	K
OFF	115	J3A	J11B	-	X	X	X	C	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X
		J4B		X	X	X	X	X	-	X	X	-	X	X	C	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X
		J4K		X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X	C	X	X	X	X	X	X
		J4L		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X
		J4C		X	X	X	X	X	X	X	X	-	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X
		J4J		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	-	X	X	X	X	X	C	X	X	X	X	X	X
ON	115	J3B		X	-	X	X	X	X	X	X	X	C	X	X	X	X	X	C	X	X	X	X	C	X	X	X	X	X	X	X
		J3C		X	X	-	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ON	230	J3C		X	X	-	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
		J3D		X	X	X	-	X	X	X	-	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
		J4B		X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
		J4C		X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X
		J4J		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	-	X	X	X	X	X	X	X	X	X	X	X	X
		J4L		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
		J12A		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ON	28	J11C		X	X	X	X	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
		J4B		X	X	X	X	X	X	X	X	X	X	-	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
		J12F		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	-	X	C

X = open circuit; C = less than .1 ohm; - = no test; TP = test probe

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TABLE VII Interconnecting box, J-4024/U, red compartment continuity test.

J-4024/U Switched		Low TP	High TP																							
			(-----J2------)					(--J10--)			(--J5--)			(--J6--)			(--J7--)			(--J8--)			(--P14--)		
S3	S4		A	B	C	D	F	7	15	8	A	B	C	A	B	C	A	B	C	A	B	C	A	B	E
1	OFF	P14A	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	-	X	X
		P14B	X	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	-	X
		P14E	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	-
		J2F	X	X	X	X	-	X	X	X	C	X	X	C	X	X	C	X	X	C	X	X	X	X	X
		J2B	X	-	X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	C	X	X	X	X
		J2A	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	C	X	X	X
		J2D	X	X	X	-	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X	X
		J2C	X	X	-	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X	X	X
2	OFF	J2C	X	X	-	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X
		J2D	X	X	X	-	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X	X	X	X
1	RCV1	J2B	X	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X
		J2A	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X
		J2D	X	X	X	-	X	X	X	X	C	X	X	X	X	X	X	C	X	X	X	X	X	X	X
		J2C	X	X	-	X	X	X	X	X	X	X	C	X	X	X	X	X	C	X	X	X	X	X	X
1	RCV2	J2B	X	-	X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X
		J2A	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	C	X	X	X	X	X	X
		J2D	X	X	X	-	X	X	X	X	X	C	X	X	X	X	X	X	X	X	C	X	X	X	X
		J2C	X	X	-	X	X	X	X	X	X	X	C	X	X	X	X	X	X	X	X	C	X	X	X

X = Open circuit; C = Less than .1 ohm; - = No test; TP = Test probe

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4.7.5 Electrical preconditioning. (See 3.12) Each J-4024/U shall be subjected to an electrical preconditioning (burn-in) in accordance with 3.12. Electrical preconditioning shall be accomplished as follows:

4.7.5.1 Bounce preconditioning. Each unit shall be subjected to a sixty (60) second bounce preconditioning in accordance with the following:

4.7.5.1.1 Test setup.

a. Configure test fixture on the transportation simulator (TS) such that the UUT (Unit Under Test) movement is restricted to 2 +1/-.5 inch on the end walls.

b. Configure the TS for circular synchronous motion.

c. Load the UUT onto the TS in its normal operating position.

d. Operate the TS increasing speed until a 1/32 thick shim can be inserted under one (1) edge to a distance of three (3) inches as the unit bounces.

e. Adjust timer for 60 +5/-0 seconds.

f. Turn off TS. This completes setup

4.7.5.1.2 Test procedure.

a. Load UUT in the normal operating position.

b. Operate TS, activate timer.

c. Turn off TS, remove UUT. The UUT may not be reworked, repaired, cleaned, aligned, etc. at this time.

d. Release UUT to burn-in.

4.7.5.2 Electrical preconditioning (burn-in).

a. Each unit produced shall be subjected to an electrical preconditioning period of twelve (12) failure free hours in the following environment:

(1). Operating environment, tests shall be conducted at ambient temperature, in normal humidity and atmospheric pressure.

(2). Input power, the unit shall be supplied with 22 - 32 Volts DC at connector J11 (plus to pin A, minus to pin C, and common to pin B).

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(3). Power on/off cycle, the unit shall be powered "on" continuously for twelve (12) hours with a signal applied.

(4). Electrical operational and Data rate operational verification shall be performed after the twelve (12) hour preconditioning period.

b. Burn-In Setup.

(1). Configure J-4024/U UUT's as shown in FIGURE 2.

(2). Set UUT switch settings to the positions indicated:

S1 to Power "ON"
S2 to Power Source 28 VDC
S3 to Transmit Position 1
S4 to Home Copy OFF

(3). This completes the preconditioning setup.

c. Test Procedure

(1). Connect each UUT as shown in FIGURE 3 test fixture switch S2 in the RCV1 position.

(2). Power up all burn-in rack test equipment.

(3). Observe the display terminal to see the test message (the quick brown fox jumped over the lazy dog's back) was routed through the J-4024/U without error. Leave all units "on".

(4). This completes UUT verification.

(5). After twelve (12) hours minimum preconditioning, turn off rack power.

(6). Connect each UUT as shown in FIGURE 3 and replace test fixture switch S2 in the RCV1 position.

(7). Observe the display terminal for each UUT to see that the test message (the quick brown fox jumped over the lazy dog's back) was routed through the J-4024/U without error.

(8). Remove power from the rack and disconnect the UUTs.

(9). For each UUT, perform Electrical operation verification per para 4.7.6.

(10). This completes the preconditioning.

4.7.6 Operation verification tests. Each MK-2488/G shall be subjected to an electrical operation verification test after

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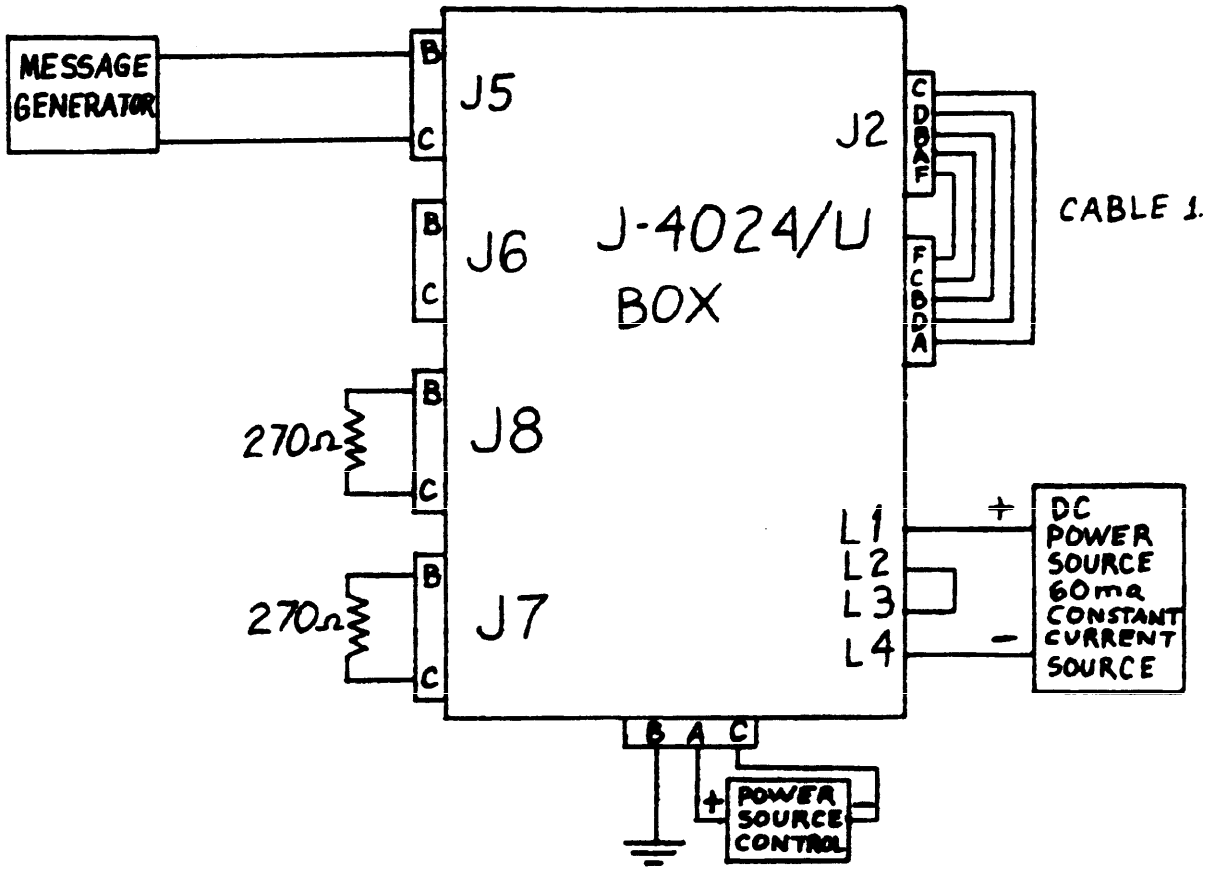


FIGURE 2 Burn - in exercise

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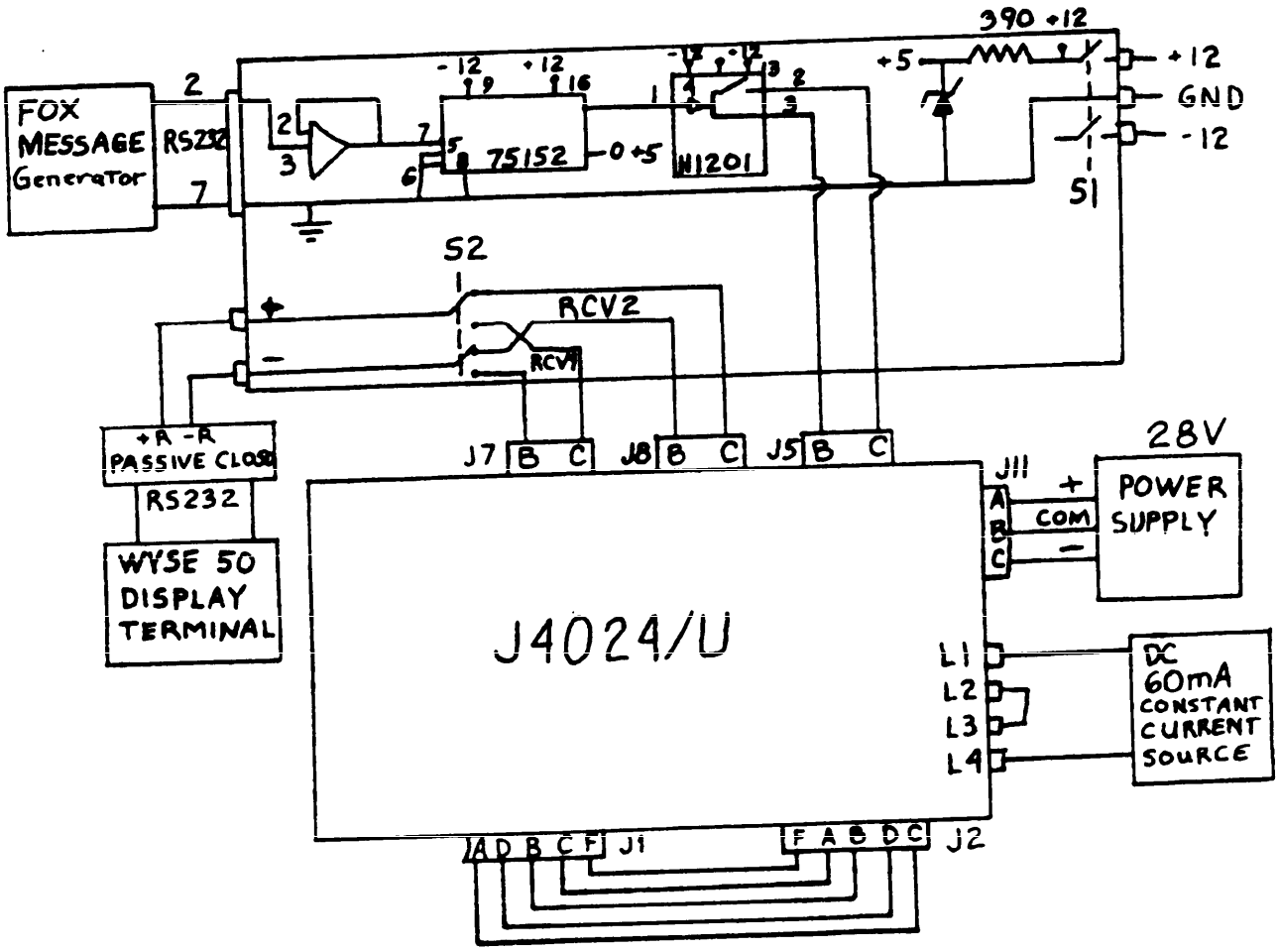


FIGURE 3 Data rate operational verification

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electrical preconditioning. Additionally, each Group B unit shall be resubjected to an electrical operation verification test after Group B testing. First Article and Group C units shall be subjected to the MK-2488/G Post EV test as required by the applicable Group C test (See 3.8).

4.7.6.1 Electrical operation verification test (group A and B units). Each J-4024/U Interconnecting Box Assembly shall be subjected to an electrical operation verification test. The electrical operation verification test shall be conducted as follows:

a. Test equipment set-up and checks. (See FIGURE 4.

(1). Test Equipment Required (or Equivalent):

1 ea Power Supply, Power Designs TW5005W
1 ea Power Supply, Power Designs 6050C
1 ea Power Supply, Powertron 350S
2 ea Ammeter, Fluke 8050A
1 ea Test Fixture (Figure 4)

(2). The 60 milliamp current must be set as follows before testing is begun:

(a). Connect the equipment as shown in FIGURE 4 with a current meter in series with the 60 milliamp current source.

(b). Set the current source power supply for an output for 12VDC and turn the current limiting knob fully CCW.

(c). Adjust power supply number 2 to 28 VDC.

(d). On the UUT turn S3 to TTY1, power to OFF, power source to 28 VDC and Home Copy to OFF.

(e). On the test fixture close S1 and switch 115/230 to 230.

(f). Adjust the powertron for an output of 220 +/- 1 VAC at 60 Hz. Power on all test equipment and then turn UUT power to ON.

(g). Adjust the current knob on power supply number 1 clockwise until the current meter in series with the it reads 60 mA.

(h). This completes the setup of the current source.

b. Test procedure.

(1). After completion of the initial setup ammeter 1 and 2 should read as follows:

(a). Ammeter 1: 0.0 to 1.0 milliamps
(b). Ammeter 2: 0.0 to 1.0 milliamps

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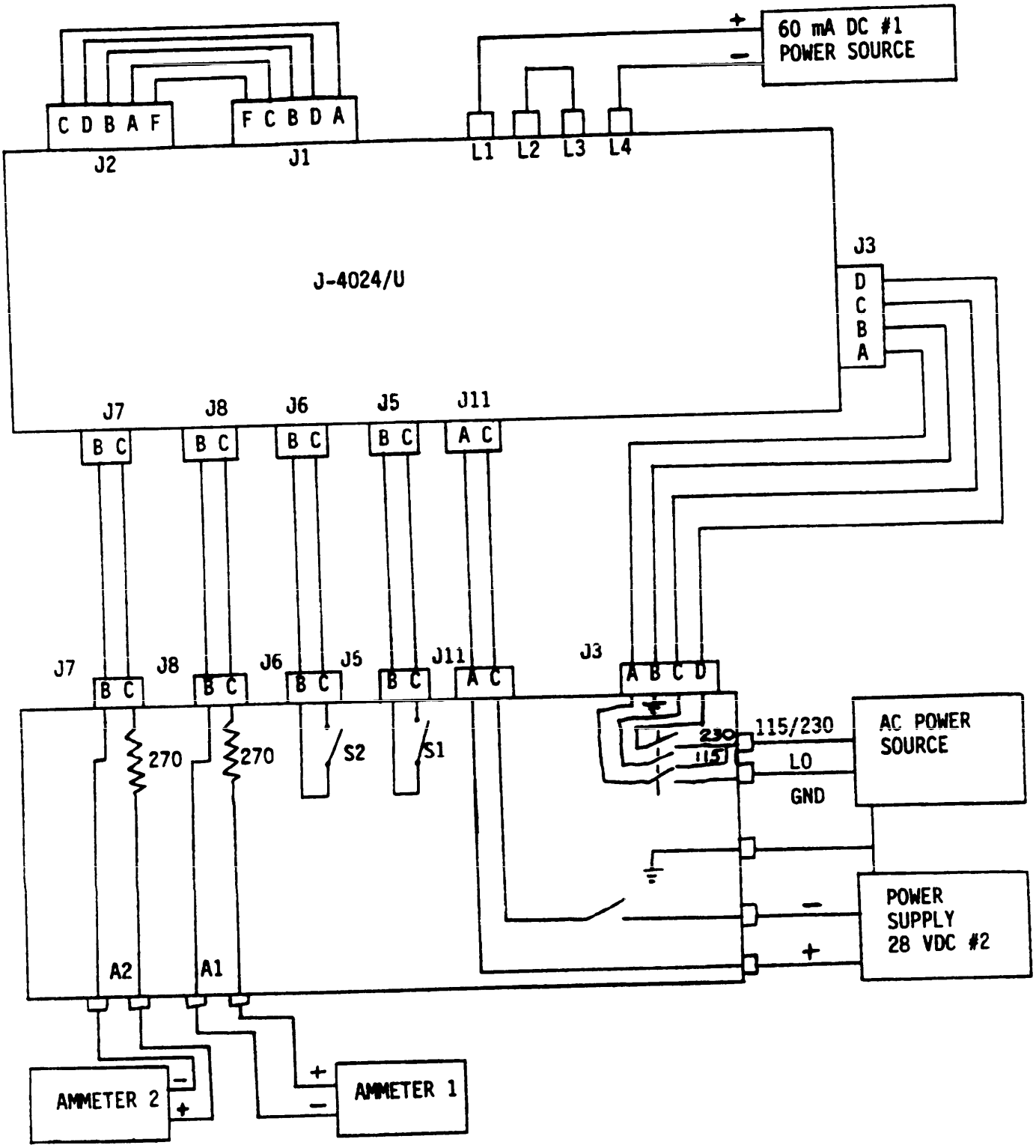


FIGURE 4 Electrical operational verification, J-4024/U

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(2). Switch S1 of the test fixture to OPEN. Measure and record the current through ammeters 1 and 2. It should be within the limits shown below.

(a). Ammeter 1: 20 +2/-1 milliamps
(b). Ammeter 2: 20 +2/-1 milliamps

(3). Turn UUT power to OFF, power source to 230. Set S1 of the test fixture to the OPEN position. Turn on UUT power, record the measurements of ammeter 1 and 2 within the following limits:

(a). Ammeter 1: 20 +2/-1 milliamps
(b). Ammeter 2: 20 +2/-1 milliamps

(4). Switch test fixture switch S1 to CLOSED. Record the measurements of ammeters 1 and 2 within the following limits:

(a). Ammeter 1: 0.0 to 1.0 milliamp
(b). Ammeter 2: 0.0 to 1.0 milliamp

(5). Turn UUT and fixture power to OFF. Turn 115/230 switch on the test fixture to 115. Adjust the powertron for an output of 115 +/- 1 VAC at 60 Hz. Set UUT power to 115.

(6) Turn fixture power to ON, S1 to OPEN and then turn on UUT power. Record the measurements of ammeters 1 and 2 within the following limits:

(a). Ammeter 1: 20 +2/-1 milliamps
(b). Ammeter 2: 20 +2/-1 milliamps

(7). Close test fixture switch S1. Record the measurements of ammeters 1 and 2 within the following limits:

(a). Ammeter 1: 0.0 to 1.0 milliamp
(b). Ammeter 2: 0.0 to 1.0 milliamp

(8). Turn the J-4024/U transmitter switch to transmitter 2. Set switch S2 of the test fixture to the OPEN position. Record the measurements of ammeters 1 and 2 within the following limits:

(a). Ammeter 1: 20 +2/-1 milliamps
(b). Ammeter 2: 20 +2/-1 milliamps

(9). Set S2 of the test fixture to the CLOSED position. Record the measurements of ammeters 1 and 2 within the following limits:

(a). Ammeter 1: 0.0 to 1.0 milliamp
(b). Ammeter 2: 0.0 to 1.0 milliamp

(10). Turn the J-4024/U power to OFF and remove cable 1 from

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the UUT. Set the J-4024/U Home Copy switch to REC 1 and turn on the J-box power. Record the measurement of the ammeter 2 within the following limit.

(a). Ammeter 1: N/A
(b). Ammeter 2: 20 +2/-1 milliamps

(11). Set J-box Home Copy switch to REC 2. Record the measurements of ammeter 1 within the following limit:

(a). Ammeter 1: 20 +2/-1 milliamps
(b). Ammeter 2: N/A

(12). Turn off UUT and fixture power before disconnecting the UUT.

4.7.6.2 MK-2488/G post EV test (first article and group C units). Each MK-2488/G (and ancillary equipment, if any) shall be subjected to the MK-2488/G Post EV Test and/or portions thereof as specified in the Individual Service Conditions requirements of 3.8.1 through 3.8.4. The MK-2488/G Post EV test shall be comprised of four subtests as follows:

a. External visual/mechanical inspection (VMI)

b. Internal visual/mechanical inspection (VMI)

c. Normal electrical operation verification (during Service Condition test)

d. Electrical operation verification (post Service Condition test).

4.7.6.2.1 MK-2488/G post EV test-external VMI. Each MK-2488/G shall be subjected to the MK-2488/G Post EV test-external VMI or applicable portions thereof as the first set of test steps to be conducted in accordance with the post service condition test requirements specified in 3.8.1 through 3.8.4. The MK-2488/G Post EV test-external VMI shall be conducted as follows:

a. Examine the J-4024/U and Mounting Base MT-6442/G (and Resilient Mount MT-6444/VSC-3 or Intermediate Mounting Base MT-6445/VSC-2 if applicable). There shall be no damage to mounting hardware and mounting mechanical connections shall be secure and shall function properly.

b. Examine the MK-2488/G Cable Assemblies. There shall be no damage to the Cable Assemblies insulation or connectors, the Cable Assemblies shall be securely connected to the J-4024/U mating connectors, and Cable Assemblies mechanical operation shall be free of binding, jaming and damage to the pins and sockets.

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c. Examine the surfaces of the J-4024/U and mounting provisions as applicable. There shall be no cracks or breaks. The finish and markings shall be free of flaking or peeling and markings shall be legible after wiping.

d. Examine the connectors on the rear panel of the J-4024/U. Connectors shall be securely attached and free of damage; connector pins and sockets shall be free of damage or bending. Connector covers shall operate properly.

e. Examine the J-4024/U fuse holders, switches and power light. The switches, switch knobs, fuse holders and power light shall be securely fastened and free of damage. The switches and fuse holders shall operate properly.

4.7.6.2.2 MK-2488/G post EV test-internal VMI. Each J-4024/U Interconnecting Box shall be subjected to the MK-2488/G Post EV test-internal VMI as the second set of test steps to be conducted in accordance with the post service condition test requirements specified in 3.8.1 through 3.8.4. The MK-2488/G Post EV test-internal VMI shall be conducted as follows:

a. Remove and examine the J-4024/U cover. The cover screws shall move freely in a line perpendicular to the cover. The cover shall not be bent or warped to the extent that it cannot be replaced on the J-4024/U housing. The EMI/RFI gasket shall be free of damage, securely attached to the cover and show no evidence of elongation or shrinkage.

b. Examine each of the three compartments of the J-4024/U. There shall be no loose or free hardware. All connectors, switches, fuse holder and power light shall be securely attached. There shall be no evidence of loose, broken or damaged solder connections to the switches, fuses, power light or connector pins.

c. Examine the printed wiring assemblies and power supply assembly in the J-4024/U. All the assemblies shall be securely mounted in the J-4024/U. All mounting provisions shall function properly. Remove all the assemblies. There shall be no evidence of physical or heat damage to the printed wiring assemblies or the Power Supply Assembly. There shall be no loose or broken mounting hardware.

d. Examine the components of the assemblies. There shall be no evidence of physical or heat damage to the components. Solder connections shall be free of cracking or other physical damage. There shall be no reduction in the seperation of component leads to the extent of creating a potential short circuit.

e. Examine the harnesses. There shall be no evidence of physical or heat damage. Shrink tubing shall be in place and there shall be no evidence of conductor nicking or other insulation damage. Harnesses shall be securely attached to the housing.

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f. Reinstall the printed wiring assemblies and Power Supply Assembly. The mounting provisions of the printed wiring assemblies and Power Supply Assembly shall function properly.

4.7.6.2.3 MK-2488/G post EV test-normal electrical operation verification (during service condition test). Each J-4024/U Interconnecting Box shall be subjected to the MK-2488/G Post EV test-normal electrical operation verification as specified in the service condition requirements in 3.8.1.1, 3.8.1.2, 3.8.1.3 and 3.8.1.5 only. The MK-2488/G Post EV test-normal electrical operation verification shall be conducted as follows:

a. Test equipment set up and checks (FIGURE 4) perform steps 4.7.6.1.a(2)(a) through 4.7.6.1.a(2)(h).

b. Steady State Checks: Before putting the J-4024/U into the test chamber perform steps 4.7.6.1.b(1) and 4.7.6.1.b(2). Leave the J-4024/U POWER switch ON.

c. Dynamic Tests: Perform steps 4.7.6.1.b(1) and 4.7.6.1.b(2) when Normal electrical operation verification is called for during the environmental test.

4.7.6.2.4 MK-2488/G post EV test-electrical operation verification (post service condition test). Each J-4024/U Interconnecting Box Assembly shall be subjected to the MK-2488/G Post EV test-electrical operation verification test as the last set of steps to be conducted in accordance with the post service condition test requirements of 3.8.1 through 3.8.4. The MK-2488/G Post EV test-electrical operation verification is identical to the Electrical operation verification test, 4.7.6.1.

4.8 Service conditions tests.

4.8.1 Installation kit, electronic equipment, MK-2488/G: mounting base, intermediate, MT-6445/VSC-2: mount, resilient, MT-6444/VSC-3: and adapter, signal, MX-10735/G.

4.8.1.1 Temperature.

4.8.1.1.1 Low temperature. (See 3.8.1.1.1) The equipment shall be subjected to the test of Method 502.2, Procedure I of MIL-STD-810. Lower the internal chamber temperature in Step 2 to -51 degrees C (-60 degrees F) for a period not less than 2 hours following stabilization of the equipment or until 24 hours have elapsed, which ever occurs first. The chamber temperature in Step 1 of Procedure II shall be adjusted to 0 degrees C (32 degrees F). In Step 5 of Procedure I perform Procedure II - operation (paragraph II-3.2 of Method 502.2). The equipment shall meet the requirements of 3.8.1.1.1.

4.8.1.1.2 High temperature. (See 3.8.1.1.2) The equipment shall be

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subjected to the test of Method 501.2, Procedure II of MIL-STD-810 for cycling exposure. In Step 12 raise the internal chamber temperature to 49 degrees C (120 degrees F). Maintain chamber temperature for 6 hours at 49 degrees C (120 degrees F). Raise the internal chamber temperature to 71 degrees C (160 degrees F) within a time period of one hour and then maintain at that temperature for four additional hours. Lower the internal chamber temperature to 49 degrees C (120 degrees F) within a time period of one hour. Repeat the above steps two additional times to make a total of three 12-hour cycles. The equipment shall meet the requirements of 3.8.1.1.2.

4.8.1.2 Low pressure (altitude). (See 3.8.1.2) The equipment shall be subjected to the test of Method 500.2, Procedure I of MIL-STD-810. For Step 2 adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level) at a rate not to exceed 2,000 fpm. The equipment shall also be subjected to the test of Method 500.2, Procedure II of MIL-STD-810. The chamber pressure in Step 2 shall be adjusted to 429.1 mm of Hg (16.9 inches of Hg or 15,000 feet above sea level) at a rate not to exceed 2,000 fpm. The equipment shall meet the requirements of 3.8.1.2.

4.8.1.3 Humidity. (See 3.8.1.3) The equipment shall be subjected to the test of Method 507.2, Procedure III of MIL-STD-810. The equipment shall meet the requirements of 3.8.1.3.

4.8.1.4 Fungus. (See 3.8.1.4) The equipment in the assembled and as ready for delivery condition, or representative samples subject to the approval of the procuring agency, shall be subjected to the test of Method 508.3 of MIL-STD-810. After 28 days, based on visual examination, the equipment shall met the requirements of 3.8.1.4. The equipment shall fail if it shows more than the growth specified in 3.8.1.4.

4.8.1.5 Sand and dust. (See 3.8.1.5) The equipment shall be subjected to the test of Method 510.2, Procedure I of MIL-STD-810. The air velocity in Step 2 shall be 1,750 +/- 200 feet per minute. In Step 5 reduce the air velocity to 300 +/- 250 feet per minute and adjust the air temperature to 63 degrees C (145 degrees F). The equipment shall be reoriented periodically to equally expose all sides of the equipment to the dust stream. Equipment shall be operated before Step 1 and during Step 11. The equipment shall meet the requirements of 3.8.1.5.

4.8.2 Interconnecting box, J-4024/U; mounting base, electrical equipment, MT-6442/G; mounting base, intermediate, MT-6445/VSC-2; and mount, resilient, MT-6444/VSC-3.

4.8.2.1 Salt fog. (See 3.8.2.1) The equipment shall be subjected to the test of Method 509.2, Procedure I, of MIL-STD-810. In Step 2 the 48 hour period shall be constant wetting. Delete step 3. Operation (mechanical and electrical) in Step 4 shall take place after a gentle wash in running water not warmer than 38 degrees C (100 degrees F) and

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drying. There shall be no corrosion of finish or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finish on metal surfaces. There shall also be no clogging or binding of moving mechanical parts. The equipment shall meet the requirements of 3.8.2.1.

4.8.2.2 Immersion. (See 3.8.2.2) The equipment shall be subjected to the test of Method 512.2, Procedure I of MIL-STD-810. After the test the equipment shall meet the requirements of 3.8.2.2.

4.8.3 Installation kit, electronic equipment MK-2488/G and adapter, signal, MX-10735/G.

4.8.3.1 Vibration. (See 3.8.3.1) The equipment as mounted in wheeled vehicles (with regular mounting base) shall be subjected to the test of Method 514.3, Procedure I for Category 8 of MIL-STD-810. The equipment shall be vibrated along each axis as follows:

5-7 Hz, 1.0 inches DA
7-92 Hz, 2.5g
92-110 Hz, 0.0058 inches DA
110-200 Hz 3.5g

The frequency of vibration shall be from 5-200 Hz and shall sweep logarithmically such that one sweep from 5-200-5 shall consume 12 minutes. The time shall be 3 hours in each of three orthogonal axes for a total of 9 hours. Following these tests, the equipment shall meet the requirements of 3.8.3.1 and shall not suffer any mechanical damage.

4.8.3.2 Shock. (See 3.8.3.2) The equipment as mounted in wheeled vehicles shall be subjected to the test of Method 516.3, Procedure I (functional shock) and Procedure VI (bench handling, without mounting base) of MIL-STD-810. When tested in accordance with Procedure I the equipment shall be subjected to shock pulses in eight axes. The -Z axis (the J-4024/U hanging from the mounting base) shall not be performed. The shock pulses of Procedure I shall be in accordance with Figure 515.3-4 for ground equipment. The equipment shall meet the requirements of 3.8.3.2 and shall not suffer any mechanical damage during or after shock tests.

4.8.4 Installation kit, electronic equipment, MK-2488/G and mount, resilient MT-6444/VSC-3.

4.8.4.1 Vibration. (See 3.8.4.1) The equipment as mounted in a tracked vehicle (with resilient mount) shall be subjected to the test of Method 514.3, Procedure I, for Category 8 of MIL-STD-810. The equipment shall be vibrated along each axis as follows:

5-5.5 Hz 1.0 inches DA
5.5-30 Hz 1.5g

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30-50 Hz 0.033 inches DA
50-500 Hz 4.2g

The frequency of vibration shall be from 5-500 Hz and shall be swept logarithmically such that one sweep from 5-500-5 shall consume 15 minutes. The time shall be 3 hours in each of the three orthogonal axes for a total of 9 hours. The equipment shall meet the requirements of 3.8.4.1 and shall suffer no mechanical damage.

4.8.4.2 Ballistic shock. (See 3.8.4.2) The equipment, as mounted in the tracked vehicle (with resilient mount), shall be subjected to the test of MIL-S-901 (for Type A, lightweight equipment, Grade A, Class 2). Mounting of the equipment on the test plate shall simulate the most severe condition which may be encountered in service. After the test, the equipment shall meet the requirements of 3.8.4.2.

4.8.5 Cable assembly, power, electrical, CX-13315/G and cable assemblies, special purpose, electrical, CX-13316/G and CX-13317/G.

4.8.5.1 Axial load. (See 3.8.5.1) Each sample Cable Assembly shall be subjected to an axial load of 250 +/- 5 lbs. applied through dummy mating connectors and requirements of 3.8.5.1 shall be met. After the axial load test the Cable Assembly under test shall be subjected to a dielectric withstanding voltage test in accordance with Method 301 of MIL-STD 202, Test Condition B, using a potential of 500 VDC on each conductor in turn, with all other conductors and the connector shell tied together.

4.8.5.2 Salt fog. (See 3.8.5.2) Each sampled Cable Assembly shall be subjected to the test of Method 509.2, Procedure I, of MIL-STD-810. In Step 2 the 48 hour period shall be constant wetting. Delete Step 3. Operation (mechanical and electrical) in Step 4 shall take place after a gentle wash in running water not warmer than 38 degrees C (100 degrees F) and drying. After the Salt fog test the Cable Assembly under test shall be subjected to a dielectric withstanding voltage test in accordance with Method 301 of MIL-STD-202 using using a potential of 1000 VDC +/- 10 VDC on each conductor in turn, with all other conductors and the connector shell tied together. Then insulation resistance shall be measured in accordance with Method 302 of MIL-STD-202, Test Condition B, using a potential of 500 VDC on each conductor in turn, with all other conductors and the connector shell tied together. The requirements of 3.8.5.2 shall be met.

4.8.5.3 Immersion. (See 3.8.5.3) Each sample Cable Assembly shall be subjected to the test Method 512.2, Procedure I of MIL-STD-810, with the KG-84 end of the Cable Assembly mated and the J-4024/U end unmated. After the immersion test the Cable Assembly under test shall be subjected to a dielectric withstanding voltage test in accordance with Method 301 of MIL-STD-202 using a potential of 1000 VDC +/- 10 VDC on each conductor in turn, with all other conductors and the

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connector shell tied together. Then insulation resistance shall be measured in accordance with Method 302 of MIL-STD-202, Test Condition B, using a potential of 500 VDC on each conductor in turn, with all other conductors and the connector shell tied together. The requirements of 3.8.5.3 shall be met.

4.8.5.4 Cold bend. (See 3.8.5.4) Each sample Cable Assembly shall be capable of being peripherally coiled over a two and one-half (2.5) inch mandrel (1 cycle) and uncoiled from the mandrel (1 cycle) for a minimum of 20 cycles. The Cable Assembly shall have been subjected to a temperature of -40 degrees C, +/-2 degrees C, for a period of not less than 20 hours prior to the cold bend test and shall be subjected to a temperature of -40 degrees C, +/-2 degrees C, during the test and with no interruption of this exposure between the 20 hour soak period and the test. The bend moment is to start as close to one end connector as practical but no closer than one quarter (1/4) inch from the solder joint between the conduit and the connector backshell. The bending shall proceed to the other end of the cable as close to the end connector as practical but no closer than one quarter (1/4) inch from the solder joint between the conduit and the backshell for tnis connector. The coiling and uncoiling process shall be accomplished with a minimum axial load of twenty-five (25) pounds applied to the cable through mating connectors engaged with the cable and connectors. The coiling process shall be accomplished at a rate of five turns per minute. After the Cold Bend test the Cable Assembly under test shall be subjected to a dielectric withstanding voltage test in accordance with Method 301 of MIL-STD-202 using a potential of 1000 VDC +/- 10 VDC on each conductor in turn, with all other conductors and the connector shell tied together. Then the insulation resistance shall be measured in accordance with Method 302 of MIL-STD-202, Test Condition B, using a potential of 500 VDC on each conductor in turn, with all other conductors and the connector shell tied together. The requirements of 3.8.5.4 shall be met.

4.9 Electromagnetic interference. (See 3.9) The MK-2488/G shall be subjected to the tests of MIL-STD-462 with Notice 3 and meet the requirements of 3.9.

4.10 Tempest. (See 3.10) The MK-2488/G shall be subjected to the tests of NACSIM 5100A and meet the requirements thereof.

4.11 Interchangeability. (See 3.6) The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect

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insertion of items.

c. Location of switches, sockets and fasteners.

d. Location of mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts or plug-in assemblies and location of the mating parts on the plug-in assemblies.

4.12 Visual and mechanical. The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (Systems Safety Engineering) which can be determined visually.

4.13 Markings. (See 3.5.1) The equipment shall be inspected to verify that the markings conform to MIL-M-13231.

4.14 Serial numbers. (See 3.5.2) The equipment shall be inspected to verify that the serial numbers appear on each Interconnecting Box, J-4024/U.

4.15 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance Inspection of Packaging.

4.15.1 First article inspection of packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation requirements of MIL-P-116.

4.15.2 Quality conformance inspection of packaging.

4.15.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.15.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with Group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified herein.

4.15.2.3 Packaging inspection. Inspection of packaging for shipment and storage shall consist of the examinations specified in Table VIII Packaging inspection provisions. Lot information shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in

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accordance with MIL-STD-105, using a single sample plan and Acceptable Quality Level of 4.0 percent defective.

TABLE VIII Packaging inspection provisions

NO.	CHARACTERISTICS	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight& Measure
108	Strapping not in accordance with specification incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. PACKAGING. (See 6.2)

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-T-45542.

5.2 Marking.

5.2.1 Levels A and B. In addition to any special or other identification marking required by the procurement document each unit pack and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The Installation Kit, Electronic Equipment, MK-2488/G, is a solid state level converter intended for use with the Dedicated Loop Encryption Device, TSEC/KG-84A when it replaces a TSEC/KW-7 in various tactical record traffic assemblages. Operating power is obtained from the assemblage power provisions.

6.2 Ordering data. The procurement document should specify the following:

a. Title, number and date of this specification and any

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amendment thereto.

b. Type and quantity of ancillary equipment (See 3.2.1).

c. Requirement for First Article Plan and reporting First Article data (See 4.3.3).

d. Deposition of Group C failures (See 4.5.3.3).

e. Level A or B preservation and packaging (See 5).

f. Special packaging and marking instructions.

g. When First Article inspection rough handling tests are not required.

h. When First Article packaging inspection requires procuring agency approval prior to production packaging.

6.3 Level B preservation. When Level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.5 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (See 4.5.3.3).

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make the maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Location of verification inspection. It is desirable that verification inspection be performed at a location that will minimize handling which might cause damage to the equipment. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Definitions.

6.8.1 Branching. Branching is a connected arrangement of filaments (hypol) formed by shoots or secondary items growing from the main stem or filament (hyphae).

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6.8.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or in a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.8.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms, when present in large numbers, may provide a colony visible to the naked eye.

6.8.4 Sporulation. Sporulation is the formation of minute unicelluar reproductive or dormant bodies, called spores.

6.8.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungi filaments. Tubular germination constitutes restricted individual growth not proceeding to colonization.

6.8.6 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of metal. The appearance resembles loose book pages or onion skin peeling.

6.9 Subject term (keyword) listing.

Installation kit
MK-2488/G
KG-84 installation kit
Comsec
KG-84
J-4024/U
MT-6442/G
MT-6444/VSC-3
MT-6445/VSC-2
MX-10735/G
CX-13315/G
CX-13316/G
CX-13317/G

Custodian
Army-CR

Preparing Activity
Army-CR
Project 5815-A121

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-49287 (CR)	Installation Kit, Electronic Equipment, MK-2488/G

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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